

A Non-Proximal Firmly Nonexpansive Expected-Risk Minimizer

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Status

This packet gives a candidate counterexample, likely valid, to the universal form of a future-work question in Bredies–Chirinos–Rodriguez–Naldi [1]. It does not contradict their numerical experiments. It shows that extra statistical or structural assumptions are needed if one wants a theorem forcing learned firmly nonexpansive maps to be proximity operators.

Source Question

The source paper studies the expected-risk problem

$$N^* \in \arg \min_{N \in \mathcal{N}} F(N), \quad F(N) = \int_{\mathcal{X} \times \mathcal{X}} \|N(\bar{x}) - \bar{y}\|^2 d\mu(\bar{x}, \bar{y}),$$

where $\mathcal{N}$ is the class of nonexpansive maps. The corresponding firmly nonexpansive learned operator is obtained by the change of variables $T = (\text{Id} + N)/2$.

Near the end of the “Third experiment: proximity operators” discussion, the authors ask whether learned firmly nonexpansive operators preferentially converge to proximity operators, or, more strongly, whether proximity operators are the only objects in the argmin of this expected-risk problem for “real” data distributions μ .

learned operator, over all the 150 tested paths γ . Right: histogram of values $\text{NormCirc}(T; \gamma)$ for the 150 tested paths.

We compared with the family

$$T_\alpha(x) = \alpha R_{90}x + (1 - \alpha)x, \quad R_{90} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix},$$

which linearly blends a pure rotation (non-conservative) with the identity (the proximity operator of the zero function, and clearly a gradient).

We compute the mean of NormCirc over all tested circles for both the learned operator and T_α , and define α^* to be the tested value of α such that NormCirc defined above for T_α and the learned operator T are the most similar. In this way, we can think of α^* as quantifying in some sense the amount of rotational content of the learned operator; smaller α^* means “more gradient-like”. The results in Figure 7 and Figure 8 show that the learned operator has a very small mean normalized circulation. We obtain approximately 3.8×10^{-3} for the operator trained with the butterfly image (Figure 7) and 4.7×10^{-4} for the operator trained with the MNIST dataset (Figure 8). This corresponds to a small α^* on the calibration curve (around 4.6×10^{-3} and 4.2×10^{-4} , respectively). The histogram of deviations around the mean further indicates that the circulation remains small across radii and centers, not just on average. Altogether, this provides strong evidence that the learned (firmly nonexpansive) operator is close to conservative, i.e., close to a gradient mapping (and thus a proximity operator) on the tested domain. This finding is noteworthy: despite not imposing circulation constraints or similar during training, the best data-fitting, firmly nonexpansive operators we obtained appear to be nearly gradient maps. This raises an interesting question for future work: do data-driven, firmly nonexpansive operators preferentially converge to proximity operators, or, even more profoundly, are proximity operators the only objects in the argmin of (CP) for “real” data distributions μ ?

5. Conclusions. Plug-and-Play (PnP) models [88] have turned to be very successful in a wide variety of applications such as image recovery problems. In this work, considering standard statistical learning theory [32], we present a complete theoretical framework for studying

Proof Intuition

The obstruction is rotational. A proximity operator of a convex function is a conservative gradient field, hence its circulation around every closed curve is zero. But the average of the identity with a quarter-turn rotation is still firmly nonexpansive while retaining nonzero circulation. If the data are generated exactly from that map and the input distribution has full support, zero expected risk forces every minimizer to agree with the rotational map everywhere. Thus the unique learned firmly nonexpansive operator is not proximal.

Counterexample

Let

$$R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \frac{\text{Id} + R}{2}.$$

Let X be a standard Gaussian random vector in $\mathbb{R}^2$ and set $Z = TX$. Equivalently, in the paper's nonexpansive formulation, set

$$N = 2T - \text{Id} = R, \quad Y = NX = RX.$$

Let μ' be the law of (X, Z) and let μ be the law of (X, Y) . Both distributions have finite second moments and therefore satisfy the source paper's expected-risk integrability hypothesis.

Theorem 1. *For the direct firmly nonexpansive risk*

$$F'(S) = \mathbb{E}\|S(X) - Z\|^2, \quad S \text{ firmly nonexpansive,}$$

the unique minimizer is $S = T$. The minimizer T is not a proximity operator of any proper lower semicontinuous convex function on $\mathbb{R}^2$.

Proof. First, T is firmly nonexpansive. For every $h \in \mathbb{R}^2$,

$$\|Th\|^2 = \frac{1}{4}\|h + Rh\|^2 = \frac{1}{2}\|h\|^2 = \langle Th, h \rangle,$$

because R is orthogonal and skew-symmetric. This is exactly the linear firm-nonexpansiveness condition $\|Th\|^2 \leq \langle Th, h \rangle$.

Since $Z = TX$, this T has risk $F'(T) = 0$, so the minimum risk is 0. If S is another firmly nonexpansive minimizer, then $\mathbb{E}\|S(X) - TX\|^2 = 0$, hence $S(X) = TX$ almost surely. Firmly nonexpansive maps are continuous. The standard Gaussian has positive mass on every nonempty open ball in $\mathbb{R}^2$. Therefore, if $Sx \neq Tx$ at any point x , continuity would give a nonempty open ball on which S and T differ, contradicting almost-sure equality. Hence $S = T$ on all of $\mathbb{R}^2$.

It remains to show that T is not proximal. A proximity operator of a proper lower semicontinuous convex function is the gradient of a convex C^1 potential [1, 2]; in particular, its circulation around every closed C^1 curve is zero. On the unit circle $\gamma(t) = (\cos t, \sin t)$, $0 \leq t \leq 2\pi$, we have $\gamma'(t) = R\gamma(t)$ and

$$\int_{\gamma} T(x) \cdot dx = \int_0^{2\pi} \left\langle \frac{\gamma(t) + R\gamma(t)}{2}, R\gamma(t) \right\rangle dt = \int_0^{2\pi} \frac{1}{2} dt = \pi.$$

The circulation is nonzero, so T is not conservative and hence cannot be a proximity operator. $\square$

Corollary 1. *In the paper's nonexpansive formulation (CP), the distribution $\mu = \mathcal{L}(X, RX)$ has the unique minimizer $N = R$. The associated learned firmly nonexpansive operator $(\text{Id} + N)/2$ is non-proximal.*

Proof. The rotation R is nonexpansive and has zero risk for (CP). The same full-support argument used above shows that any nonexpansive zero-risk minimizer must agree with R everywhere. The associated firmly nonexpansive map is $(\text{Id} + R)/2 = T$, which is non-proximal by the theorem. $\square$

Verification Notes

The proof is analytic. The included script `code/verify_linear_counterexample.py` checks:

- the matrix inequality $T^*T \leq (T + T^*)/2$;
- $\|2T - \text{Id}\| = 1$, so the corresponding (CP) variable is nonexpansive;
- the unit-circle circulation is numerically π .

Command used:

```
conda run --no-capture-output -n sandbox python code/verify_linear_counterexample.py.
```

Novelty Check and Limitations

Local indexes searched: `registry_index.tsv`, `solutions/index.tsv`, `attempts/index.tsv`, and `proof_gaps/index.tsv`, using 2407.14156, Learning Firmly, proximal argmin, nonproximal, and close firmly-nonexpansive/proximity phrases. No duplicate packet or attempt was found.

External web search on 2026-07-19 for the exact phrase “are proximity operators the only objects in the argmin”, together with close variants involving “Learning Firmly Nonexpansive Operators”, “firmly nonexpansive”, “proximity operators”, and “expected risk”, found the source paper and general plug-and-play/proximity-operator references, but no exact answer to this future-work question.

The result refutes only the universal form of the question. It does not rule out the softer “preferential convergence” possibility, nor positive theorems for natural-image distributions, separable gradient data, symmetry assumptions, or additional constraints that enforce conservativity/cyclic monotonicity.

Human Review Recommendation

Reviewers should check that the source question is interpreted as a universal mathematical claim under the expected-risk framework, and that the distinction between the source paper’s nonexpansive variable N and the associated firmly nonexpansive variable $T = (\text{Id} + N)/2$ is acceptable. The proof itself reduces to standard two-dimensional Hilbert-space calculations.

References

- [1] K. Bredies, J. Chirinos-Rodriguez, and E. Naldi, *Learning Firmly Nonexpansive Operators*, arXiv:2407.14156v3, 2025.
- [2] H. H. Bauschke and P. L. Combettes, *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*, 2nd ed., Springer, 2017.